

Scenario 1:

Flight Delay Analysis

An airline tracks flight delays (in minutes) for 20 flights. Analyze the flight delays to calculate percentiles, detect outliers, and evaluate the overall distribution.

Answer :

1. Sort the data: Arrange the 20 delays from smallest to largest.
2. Find percentiles: Calculate Q1 (25th), Q2 (median/50th), and Q3 (75th) to see the spread.
3. Calculate IQR and Outliers: Subtract Q1 from Q3 to measure variability and calculate values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.
4. Check distribution: Compare mean vs median and look at spread/shape to see if it's skewed or balanced.
5. Summarize: State the typical delay range, central tendency, and highlight extreme outliers.

Scenario 2:

Employee Salary Analysis

A company wants to analyze the salary distribution of its employees to understand the central tendency and determine whether the data is skewed.

Answer:

1. Sort the data: Arrange all employee salaries from lowest to highest.
2. Find percentiles: Calculate Q1, Q2 (median), and Q3 to capture the salary spread.
3. Calculate IQR and Outliers: Subtract Q1 from Q3 to measure variability and calculate values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.
4. Check distribution: Compare mean vs median; if $\text{mean} > \text{median} \rightarrow$ right-skew (high earners), if $\text{mean} < \text{median} \rightarrow$ left-skew.
5. Summarize: Report typical salary range, central value, and presence of extreme salaries.

Scenario 3:

Product Sales Analysis

A retail store records product sales over 15 days. Create a frequency distribution table and visualize the sales data using appropriate charts.

Answer:

1. Collect and sort data: List the 15 daily sales numbers in order.
2. Frequencies: Group the sales into ranges (e.g., 0–50, 51–100, etc.) for easier comparison and count how many days fall into each interval.
3. Visualize the data:
 - a) Histogram chart: Shows how sales are distributed across intervals.
 - b) Line graph: Highlights daily sales trends over time.
4. Summarize: Note whether sales are steady, skewed toward high/low values, or show peaks.

Scenario 4:

Student Exam Performance Analysis

A school wants to analyze the exam performance of students across three subjects: Mathematics, Science, and English. How can Data Science concepts be applied to understand their performance?

Answer:

1. Data Collection & Cleaning: Gather scores of all students in the three subjects, check for missing or incorrect entries.
2. Descriptive Statistics: Calculate mean, median, mode, standard deviation to understand central tendency and spread.
3. Distribution Analysis: Plot histograms / boxplots for each subject to see if scores are symmetric, skewed, or full of outliers.
4. Correlation Analysis: Check relationships between subjects and plot heatmaps to visualize the same.

Scenario 5:

Clinical Trial for Diabetes Medication

A pharmaceutical company conducted a clinical trial with two groups: one receiving medication and the other a placebo. Perform a hypothesis test to determine the effectiveness of the medication.

Answer:

1. Define groups:
 - Group A = patients taking the new diabetes medication.
 - Group B = patients taking the placebo.
2. State hypotheses:
 - Null hypothesis (H_0): There is no significant difference in effectiveness between medication and placebo.
 - Alternative hypothesis (H_1): The medication is more effective than the placebo.
3. Choose a test - use a two-sample t-test to compare means.
4. Compute the test statistic & p-value: The p-value tells how likely this difference is if H_0 were true.
5. Decision rule:
 - If $p\text{-value} \leq 0.05 \rightarrow$ reject $H_0 \rightarrow$ conclude medication is effective.
 - If $p\text{-value} > 0.05 \rightarrow$ fail to reject $H_0 \rightarrow$ no strong evidence medication works better than placebo